

No Candidate

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

8th Semester of B. Tech. Examination (CL)

Nov 2013

CL 405 Estimation, Costing & Financial Concepts -II

Date: 27.11.2013, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Figure 5 & table 2 gives details of a steel column. Estimate the quantity of steel work. [07]
- Q - 2 (a) Workout the quantity of foundation for following items (up to plinth level) drawings given in Figure 3 & 4. Also refer Table 1. (i) Excavation in foundation (ii) P.C.C (ii) R.C.C work in column. [14]

OR

- Q - 2 (a) Workout the quantity of following items for first floor, drawings given in Figure 3. Also refer table 1. (ii) Masonry work (iii) R.C.C work (Beam, Column and slab only) [14]
- Q - 3 (a) List the important points for preparation of construction project report. [05]
- (b) Workout quantity of septic tank shown in Figure 1. [09]

OR

- Q - 3 (a) Write different stages of project. [05]
- (b) Workout the quantity of soak pit from Figure 1. [09]

SECTION - II

- Q - 4 (a) Work out quantity of following items for slab culvert shown in Figure 6. [07]
(i) Excavation (ii) Cement concrete 1:4:8 (iii) Brick work in 1:3 (iv) R.C.C work in 1:2:4
- Q - 5 Attempt any Two [14]
- (a) What do you understand by a standard rent & how it is calculated?
- (b) Explain Mortgage.
- (c) A building is situated by the side of a main road of Lucknow city on a land of 500 sq m. the built up portion in 20m x 15m. The building is first class type and provided with water supply, sanitary and electric fittings, and the age of building is 30 years. Workout valuation of property.
- Q - 6 (a) State the condition of release of first and final bill of work. [04]
- (b) Reduced levels of ground along the centre line of a proposed road from chainage 10 to chainage 20 are given below. The formation level at the 10th chainage is 107 and the road is in downward gradient of 1 in 150 up to chainage 14 and then the gradient changes to 1 in 100 downward. Formation width of road is 10 meter and side slope of banking are 2:1(Horizontal: vertical). Length of chain is 30m. [10]

Chainage (m)	10	11	12	13	14	15	16	17	18	19	20
G.L	105	105.60	105.44	105.9	105.42	104.30	105	104.1	104.62	104	103.3
Gradient	1 in 150					1 in 100					

R.L of Formation 107.0 at 10 chainage.

OR

- Q - 6 (a) Distinguish between Regular contracts and piecework [04]
- (b) Prepare the estimate for the construction of a macadam road for one kilometer length [10]
 having following specification. Formation width of road=12.0m, Average height of bank 0.6m, Side slope 2:1 (H:V), Metalled width of road =3.8 m. Soling coat of hard stone 63mm size, 20 cm thick, sides shall be provided with brick edging, wearing coat in two layers each layer 10 cm thick of stone aggregate 50 mm nominal size. Refer figure 2.

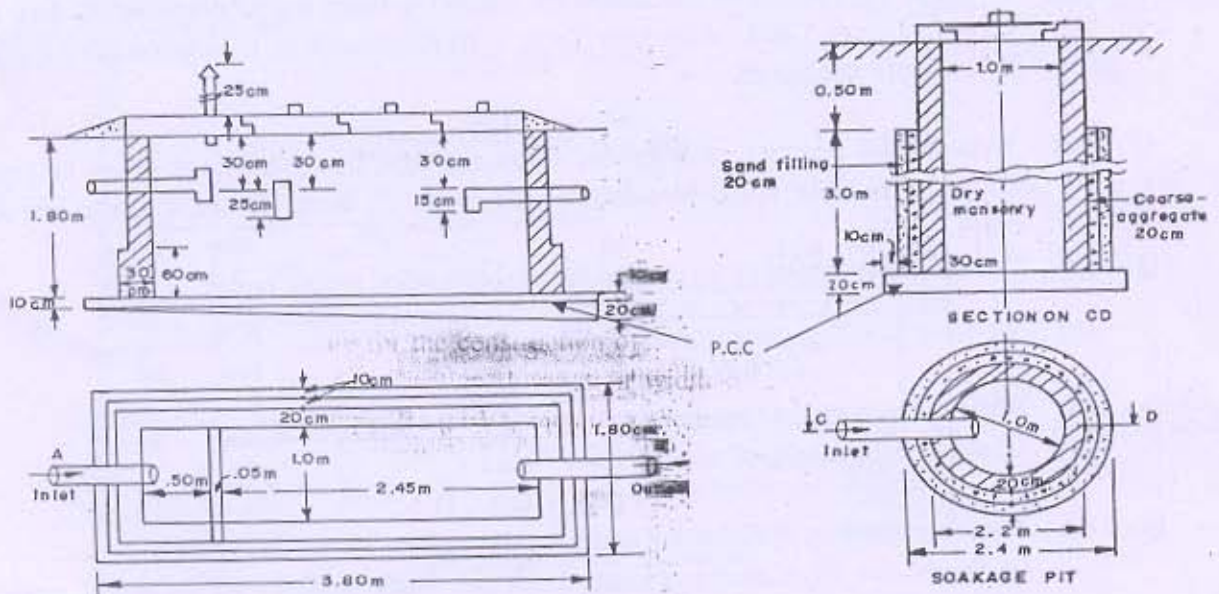


Figure 1 Q.3 (a)

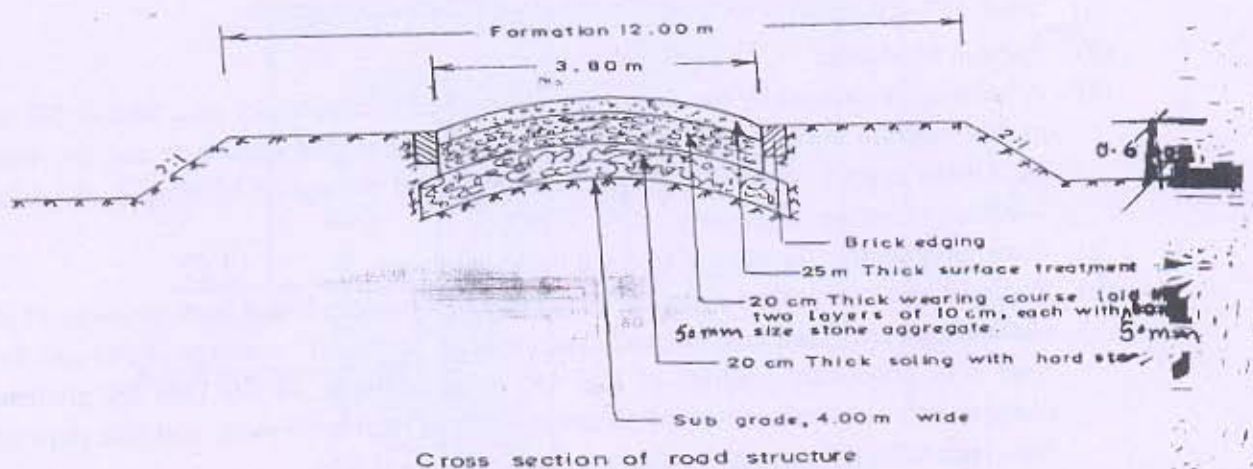


Figure 2 Q.6 (b)

Table- 1.

Column Group	C8,C9,C10, C16,C17, C18	C13	C7,C11	C14,C12, C19,C15	C5	C6	C2,C3, C4,C20	C1
Column Size	230 x 230	230 x 230	230 x 230	230 x 230	230 x 230	230 x 230	230 x 230	230 x 230
Footing size	1300 x 1300	1225 x 1225	1150 x 1150	975 x 975	900 x 900	800 x 800	725 x 725	475 x 475
Df	300	275	250	225	200	175	150	150
Df-min	150	150	150	150	150	150	150	150
Beam Group	B1 to B16 , B18 , B22 to B25				B19,B20,B21,B17			
Beam Size	230 x 380				230 x 750			

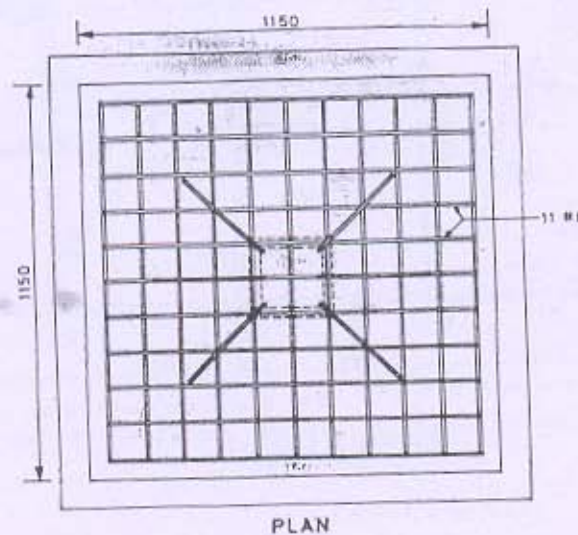
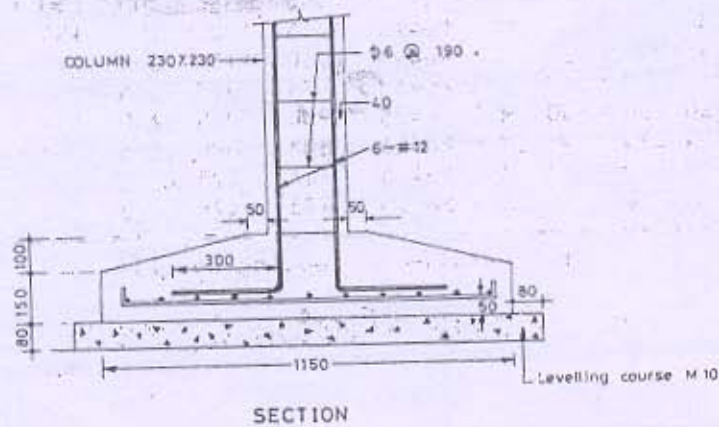


Figure 4 Q.2 (a) Column footing (column 7)

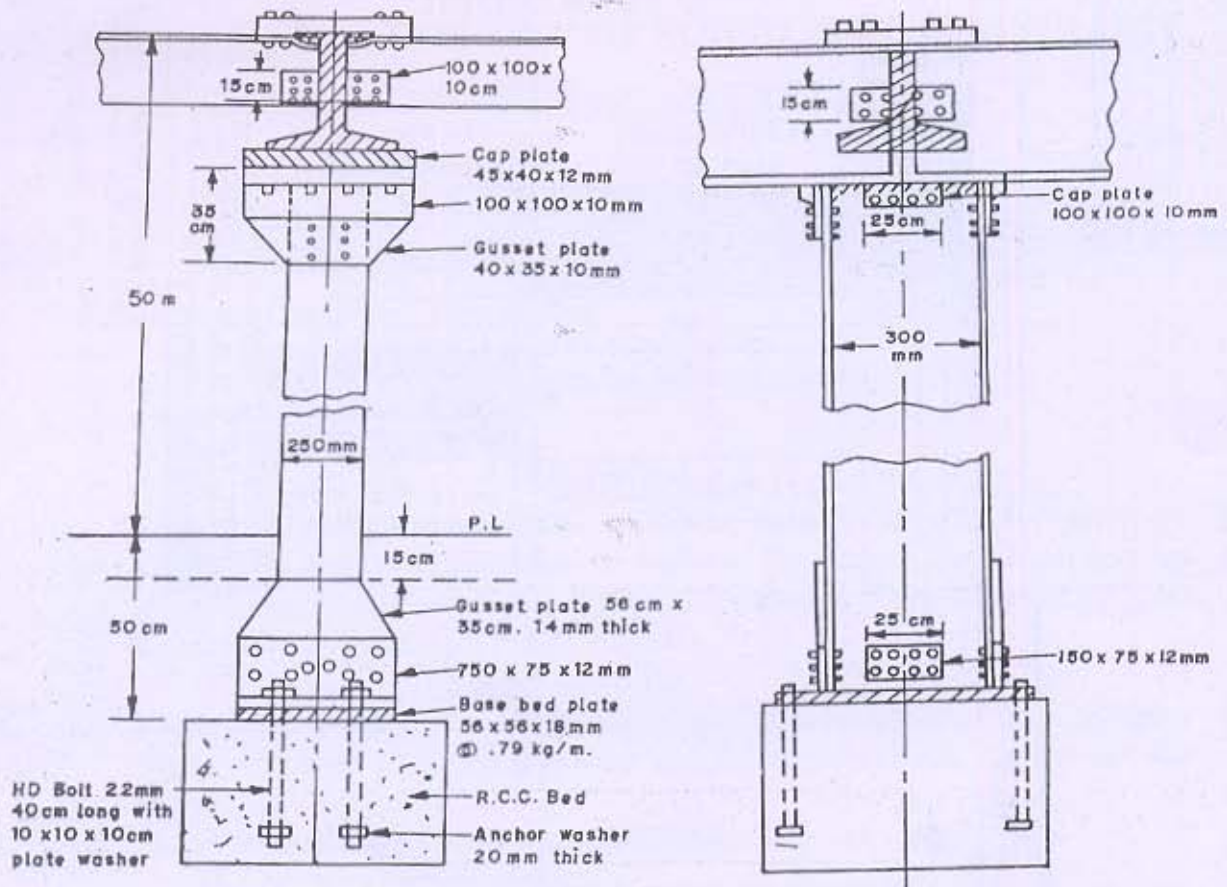


Figure 5 Q.1.

Table- 2 Weight of the sections

Section	Weight
ISHB-300	63.0kg/m
M.s plate	Weight
56 x 56 x 18	79.1
56 x 35 x 14	6.15
40 x 35 x 10	31.4
44 x 40 x 12	42.4
Cleat angle	Weight
750 x 75 x 12	20.1
150 x 75 x 12	20.1
100 x 100 x 10	14.9

